

Student Answer Script View



058-END SEMSTER-ODD-2025

Answer Sheet

Course: Computer Science and Engineering**Year/Sem:** Semester 7**18.00****Subject Name:** Computer Vision-III**50.00****Exam Date:** 18-Nov-2025

Comments submitted by student.

Q.No : 1 A)

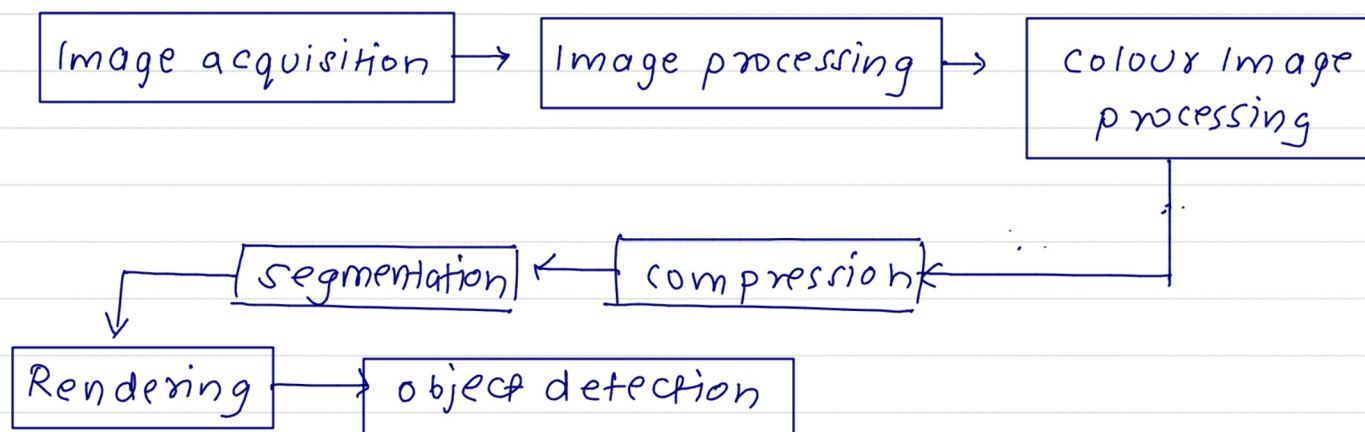
Score : 5.00 / 5.00

A drone captures aerial images of a construction site to monitor progress and detect structural cracks.

Illustrate how digital image processing steps help to extract structural details and detect possible defects.

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solution:- there are multiple steps involved in digital image processing. When we talk about aerial images of construction captured using drone we will consider the following steps of image processing



i) when the drone captures an image, an input image is acquired, and thus the image is acquired for further steps in detection

ii) Images are then processed and scaled according to the application, and here the application we are using is for crack detection, hence in the image processing step, images are scaled and transformed for a specific resolution for further steps.

iii) Then we come to the colour image processing, here the images are converted into image negative or any other colour changes are made for processing

iv) Next the images are smoothened using a filter

For example gaussian filter for a octaves of 4 until we achieved a complete smoothened image ready for detecting cracks

v) Then after the image is smoothened, its compressed and segmentation is performed to detect possible cracks on the structure.

ii) once the cracks are visible, a dedicated deep learning model uses object recognition to detect the cracks in the structure.

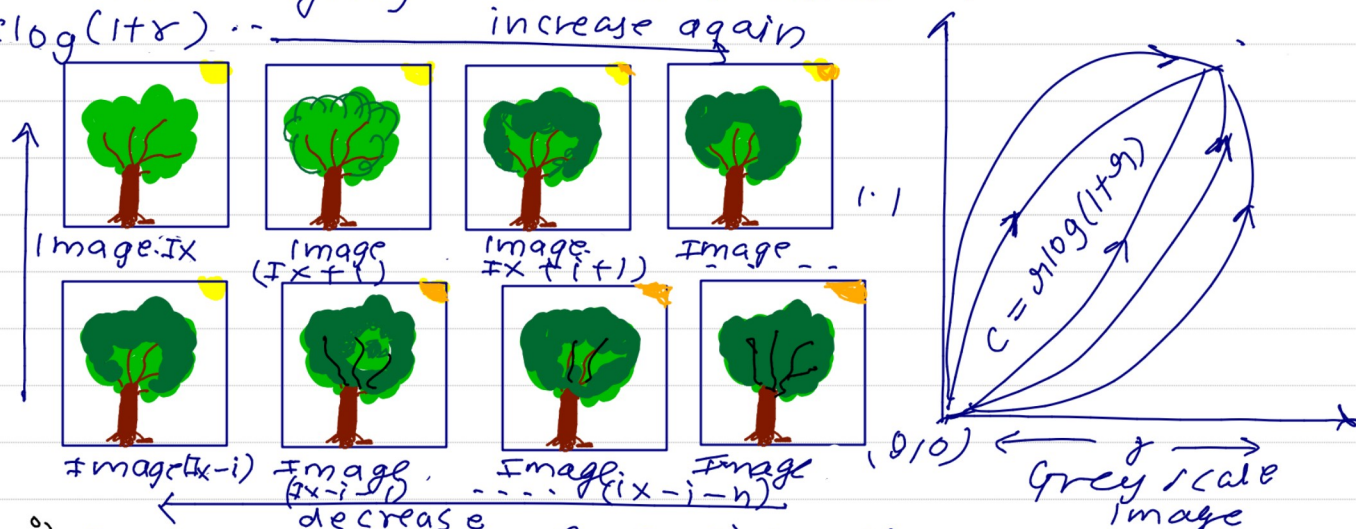
Q.No : 1 B)

Score : 0.50 / 5.00

Examine how Power-Law Transformations are used to control image contrast in photographs captured under varying illumination conditions.

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Solution:- Power-Law transformation $C = \gamma \log(1+x)$ are used to control the contrast of image under varying illumination condition $\log(1+x)$ -- increase again



i) power - Law transformations are used to control the contrast of a image captured in illumination condition

ii) As we increase the value of γ as the value of the magnitude $C = \gamma \log(1+x)$ increases as magnitude $\log(1+x)$ is a increasing function

iii) The Intensity is increased at the curves the curves hence photograph has lesser contrast and high Intensity

iv) As we move towards the edge of the graph we achieve higher contrast and vivid images as shown in analogy (1.1)

Q.No : 2 A)

Score : 5.00 / 5.00

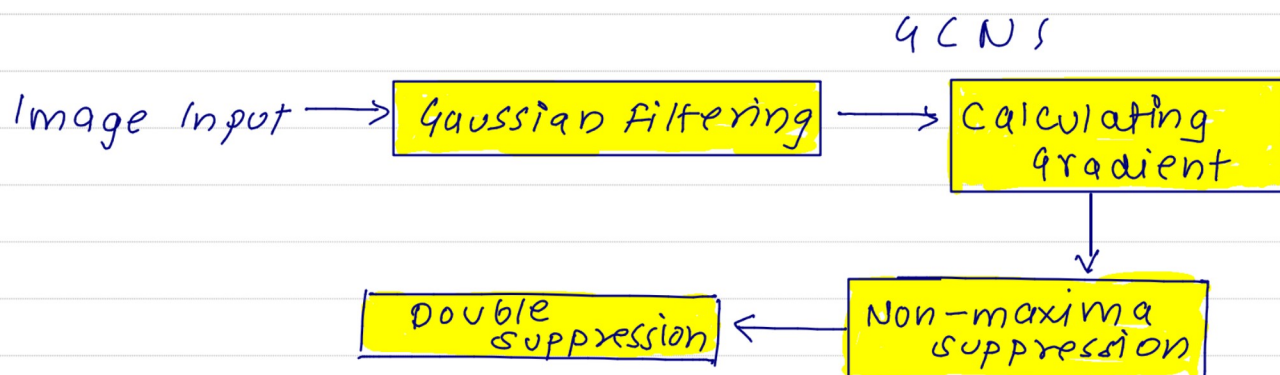
Illustrate how edge detection techniques help in obtaining important structural details from digital images.

Compare the working principles of the Laplacian and Canny edge detection methods

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Solution:- Edge detection techniques does in fact help us in obtaining structural details from a digital images

principles of canny, Edge-detection



- i) In canny Edge detection method first the images are inputted inside Gaussian filtering to smoothen the image in octaves of 4 till we get a image that is perfect for detection structural details, Even feature point is considered for further detection.
- ii) After the images are smoothened from Gaussian filtering, we calculate the gradient.

Gradient is calculated by

Gradient Formula

$$G_x = x_2 - x_1$$

$$G_y = y_2 - y_1$$

$$a = \sqrt{(G_x)^2 + (G_y)^2} \quad \text{and} \quad \theta = \tan^{-1} \left(\frac{G_y}{G_x} \right)$$

for example if the point identified here is
Let say 16

96	32	54
17	16	9
22	27	83

$$G_x = 17 - 9 = 8$$

$$G_y = 32 - 27 = 5$$

$$a = \sqrt{(8)^2 + (5)^2} = \sqrt{64 + 25} = \sqrt{89} = 9.433$$

$$\theta = 9.433 \left(\frac{5}{8} \right) = 5.89^\circ$$

ii) Non Maxima suppression:- In this process we have to take the point of feature (detected)
 - In a 16×16 image we get 8 neighbours
 9 pixel are considered, we select the point pt and rest maxima is set to 0

iv) double suppression:- if

$\| \nabla G_x \geq 0 \|$ then ≤ 0 definitely a edge

$\| \star \nabla G_x \geq 0 \|$ then $= 0$ probably a edge

$\| \nabla G_x > 0 \|$ then $+1 \geq 0$ No edge.

Q.No : 2 B)

Score : 0.50 / 5.00

Apply the concept of the Histogram of Oriented Gradients descriptor to identify and represent the shape and structure of objects within an image. Discuss with a suitable real time application

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Solution:- Histogram is a statistical representation in mathematics which can be used for several applications like determining the shape and structure of objects.

set of

- i) Let us assume we have a set of images which we are willing to represent the shape and structure
- ii) we calculate the oriented gradients descriptors of all of them, and then list them in our histogram accordingly.
- iii) once we compare all the descriptor of gradients, we can then classify each image as a function on whether its edge is in the structure or between the structure of the object.
- iv) If we have the gradient descriptor which has a lesser value, we can identify the shape and structure of the object in the image.

Q.No : 3 A)

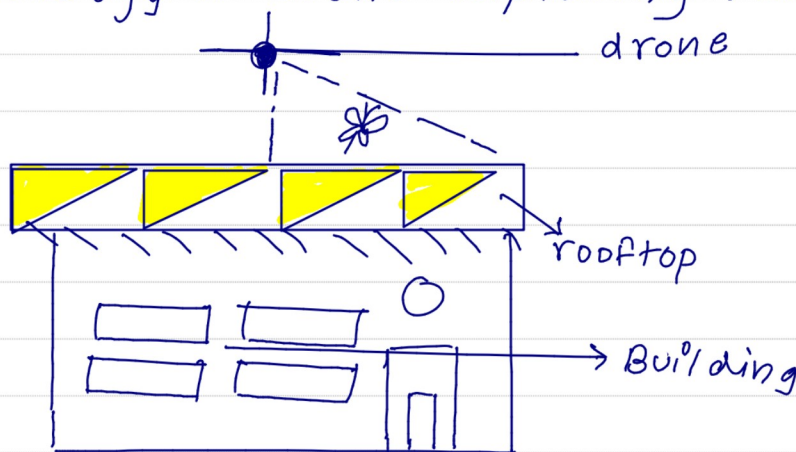
Score : 1.50 / 5.00

A drone captures rooftop images for inspection, but the photos appear misaligned and distorted due to the camera's tilt. Apply geometric transformations to adjust the tilted drone images and discuss in detail.

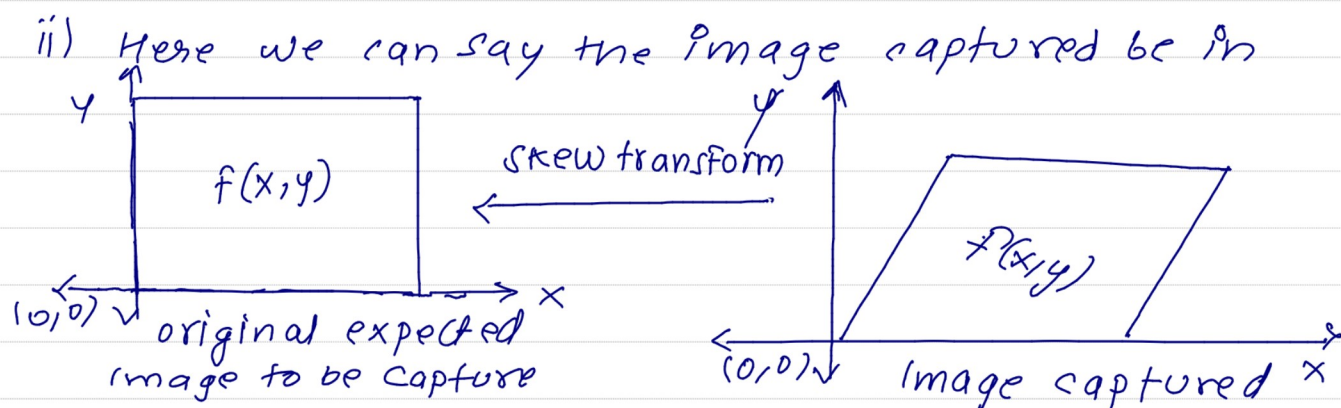
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Solution:- Analogy of drone capturing rooftop images



i) Here the drone captures image from a distance (d), we have given a situation where the image captured is tilted and not aligned but without getting compressed



ii) We can clearly use a transformation to fix it. We can apply skew geometric transformation to fix the distorted and misaligned image.

iii) Skew geometric transformations are used to re-align the distorted or misaligned image without the image losing its dimension or without compressing the image.

v) The way to do this is first we have to derive the combination to get the skew transformer

$$\begin{bmatrix} \bar{x} \\ \bar{y} \\ 1 \end{bmatrix} \begin{bmatrix} z\bar{x} \\ z\bar{y} \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & s_x \\ 0 & 1 & s_y \\ 0 & 0 & 1 \end{bmatrix}$$

$$p = \begin{bmatrix} \bar{x} \\ \bar{y} \\ 1 \end{bmatrix} = \begin{bmatrix} z\bar{x} \\ z\bar{y} \\ 1 \end{bmatrix} \dots \text{from (i)}$$

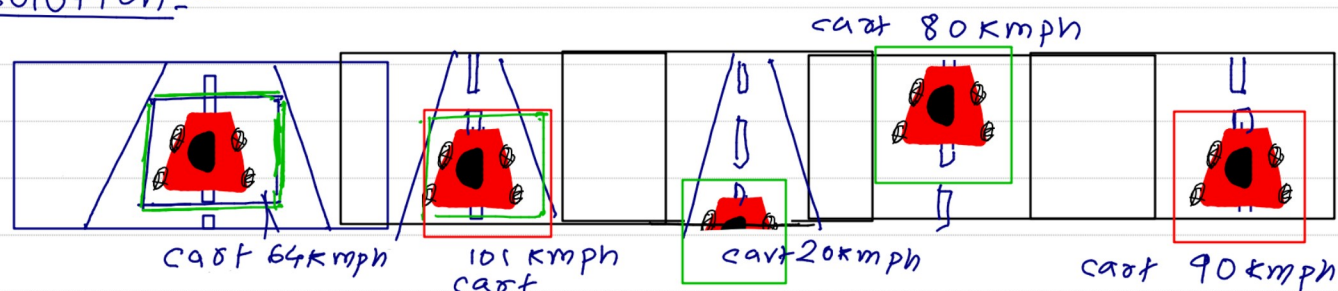
vi) We combine our skew transformer by taking the product of our misaligned image with the skew transformer to fix the image to its original shape, size and resolution -

v) Therefore, using the appropriate transformer function, we can fix the image.

Q.No : 4 A)

Score : 0.50 / 5.00

Apply the learned motion estimation framework to detect abnormal motion patterns in video streams and discuss in detail.

Solution:-

- i) Let us consider a scenario of a car in the video frame,
- ii) when the car is above the speed limit 80 it gives us a alert about the abnormal motion pattern in the video stream.
- iii) Thus patterns with overspeeding is detected and flagged using the motion estimation.

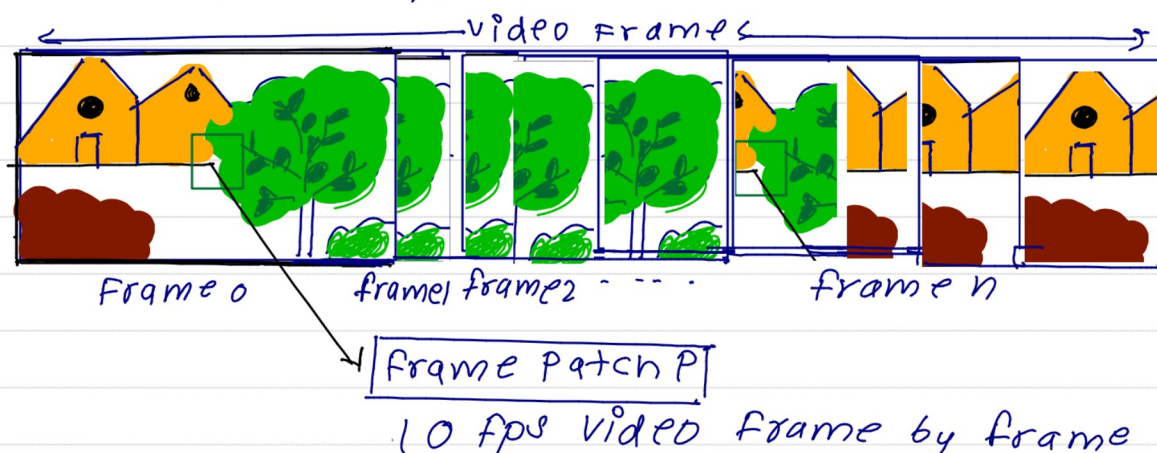
Q.No : 4 B)

Score : 2.00 / 5.00

In a video tracking application, a system needs to estimate how an object moves between consecutive frames. Utilize the Lucas-Kanade method to estimate the motion of objects across consecutive frames.

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Solution:- Let us consider the scenario of capturing a video of an object that is to be detected in an optical flow



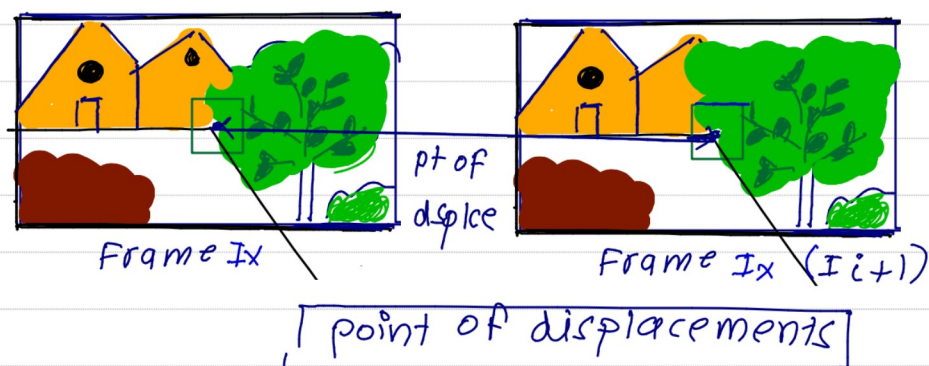
i) Here we can see we have the video frame by frame represented in this analogy

ii) In the Lucas Kanade method of estimation if there is any motion of object/optical flow we first consider a patch which is

- Not smoothened
- Does not have any distortion

iii) In the next consecutive frame $I_x(I_{i+1})$ we use the same patch as a reference to detect the presence of any type of optical flow or motion.

iv) Next we can also use sum of square distance to see any displacement or detect motion inside the patch, if found motion detected, we can then estimate the motion of objects

v_i^0 

vii) In this way we can utilize Lucas - Kanade method to estimate motion of object across consecutive frame

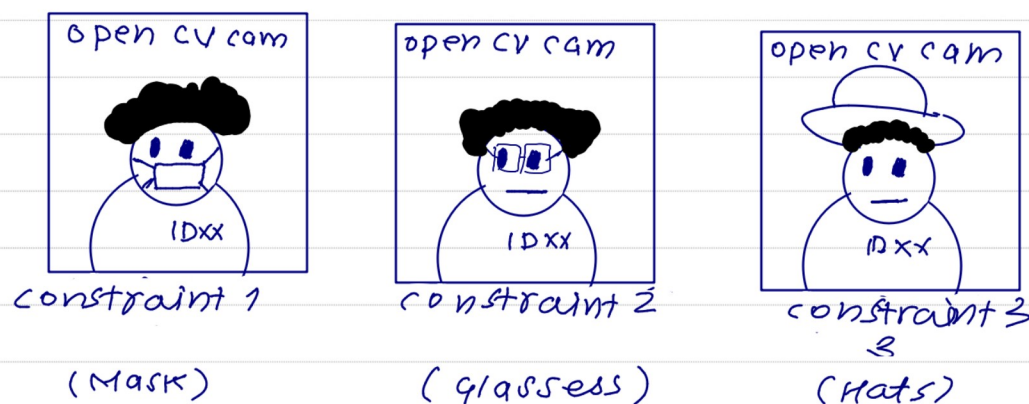
Q.No : 5 A)

Score : 2.00 / 5.00

Develop an intruder detection system using face recognition for a secure facility where individuals may wear masks, glasses, or hats and explain the recognition pipeline to create a reliable method that ensures accurate identification of unauthorized persons under varying appearance conditions.

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Solution:-

(i) To Ensure a secure and a reliable face recognition for a secure facility we have to consider mainly three test cases, which include mask, glasses and hats.

ii) If we consider the features of the users which are still visible under all test cases are eyes.

iii) Since each of the human being has a unique iris. We can use the iris of all the authorized people in organization to register and capture the iris and load the datasets of the users in the system.

iv) To detect the unique iris, we can use data clustering in machine learning to detect a unique iris of authorized member in the organization which will be trained using the dataset we fed for training.

v) We will also use open CV and yolo to detect the iris and backed with cloud infrastructure deployed machine learning model to recognize detect and classify authorized users and approve them.

vi) In this way we can implement the pipeline for keeping unauthorized users yet keeping accurate and reliable even with glasses, mask and hat on.

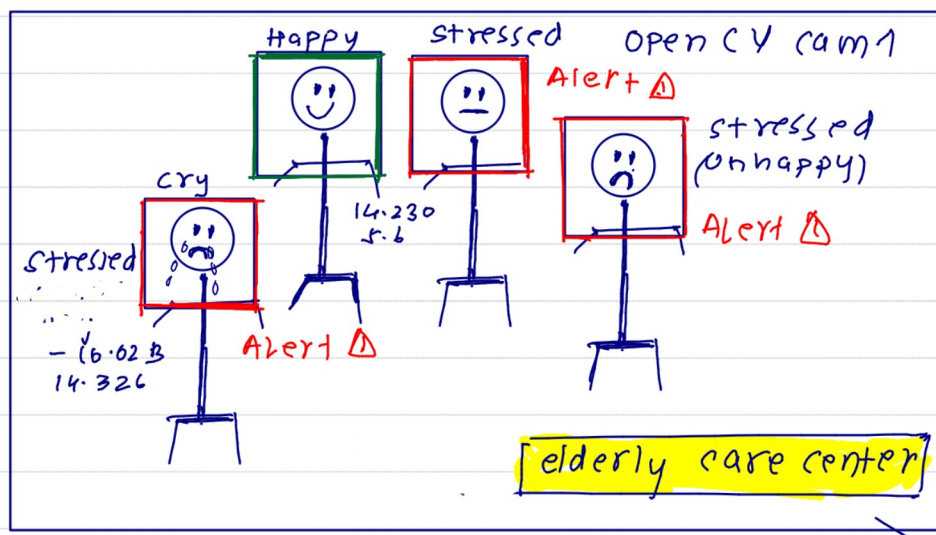
Q.No : 5 B)

Score : 0.50 / 5.00

Apply a computer vision-based activity detection model to develop an elderly care monitoring system that observes indoor movements and detects signs of falls or distress. How would you apply these techniques to automatically identify normal vs. abnormal activities and trigger alerts in real time?

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- (i) We can use application of computer of computer vision to create a elderly care monitoring system.
- ii) To implement this we need a classification model, which will classify "Happy" and classify "stressed".
- iii) When a person in the elderly day care centre is happy smiling and walking normally, we can use CNN to classify happiness or effect of normality.
- iv) The ml model can be trained with happy elder datasets and can be tested out by fine accuracy by deploying into production.
- v) Similarly stressed or abnormalities can be detected by sad and stressed adults dataset which would then classify it as abnormal activities to trigger alerts.



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